MOVIELENS RATING PREDICTION PROJECT

HarvardX: PH125.9x Data Science

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Abstract

This project implements an incremental, hierarchical predictive modeling approach to estimate movie ratings using the MovieLens 10M dataset. To ensure methodological rigor and prevent data leakage, the environment was strictly partitioned into a development dataset (edx) and an isolated validation set (final_holdout_test). An Exploratory Data Analysis (EDA) visually isolated systemic biases, revealing a general positivity skewness (global mean of 3.51), significant variations in user rating scales (user bias), and high structural sparsity. A sequence of models was progressively engineered: beginning with a baseline Naive Average ($RMSE = 1.06120$), incorporating Movie Effects ($RMSE = 0.94391$), and adding User Effects ($RMSE = 0.86535$). Ultimately, a Regularized Multilevel Model was developed to penalize high-variance estimates from low-sample outliers. This regularized architecture achieved the peak predictive performance with a final $RMSE$ of 0.86482, successfully outperforming the target benchmark threshold of 0.86490.

1 Overview

This project investigates the problem of predicting movie ratings using the MovieLens 10M dataset. The main objective is to develop and evaluate models capable of estimating how users rate movies based on historical rating patterns. The analysis follows a structured data science workflow, including data preparation, exploratory data analysis, model building, and performance evaluation.

Different approaches were compared, starting from a simple baseline model and progressing to models that incorporate movie effects, user effects, and regularization. The ultimate goal is to identify a model that achieves strong predictive performance, measured by the **Root Mean Squared Error (RMSE)**, also illustrating key concepts in recommendation systems and statistical learning.

1.1 Introduction

Recommendation systems are a fundamental application of machine learning, driving user engagement across digital platforms like streaming services and e-commerce. By analyzing historical interactions, these algorithms can predict a user's preference for an unseen item based on shared behaviors and intrinsic biases.

The **MovieLens dataset**, generated by the GroupLens Research lab at the University of Minnesota, is a classic benchmark for such tasks. In this project, the prediction of movie ratings is explored as a practical application of data science methodologies. Rather than relying on simple averages, this analysis implements collaborative filtering techniques to capture the specific nuances of both movie popularity and individual user evaluation standards.

1.2 Aim of the project

The primary goal of this project is to construct and evaluate a machine learning algorithm capable of predicting user ratings for movies using the `edx` subset of the MovieLens 10M dataset. The objective is to achieve an **RMSE lower than 0.86490** on a separate validation dataset (`final_holdout_test`).

The approach begins with a baseline global average and incrementally builds complexity by incorporating movie and user biases. Ultimately, the project seeks to determine how regularized multilevel models handle outliers and mitigate the risk of overfitting caused by users or movies with very few ratings.

Model performance is evaluated using the **Root Mean Squared Error (RMSE)**. The function that computes the RMSE for vectors of observed scores and their corresponding predictors is:

$$RMSE = \sqrt{\frac{1}{N}\sum_{u,i}(\hat{y}_{u,i} - y_{u,i})^2}$$

2 Dataset and Exploratory Data Analysis (EDA)

2.1 The Movielens Dataset

The MovieLens dataset is not merely a collection of random matrices; it carries a foundational history in the evolution of modern recommendation architectures. Developed and meticulously maintained by the GroupLens Research lab at the University of Minnesota, the platform was created as an open research environment to study collaborative filtering algorithms (Harper & Konstan, 2015)." The historical interactions within this dataset were generated by actual users of the MovieLens web-based recommendation engine (movielens.org), **a non-commercial platform where movie enthusiasts voluntarily rate films on a 0.5-to-5.0-star scale to receive personalized viewing suggestions**.

In the landscape of computer science and data analytics, the MovieLens 10M repository is universally recognized as a **premier benchmark** for predictive modeling.

Its importance stems from two critical factors:

- **Real-World Interaction Dynamics:** Unlike synthetic data, it captures **genuine human behavioral patterns**, including intrinsic user biases, varying rating thresholds, and highly complex interaction mechanics.
- **The Sparsity Standard:** It mirrors the **exact mathematical challenges** faced by major digital platforms (such as streaming services and e-commerce giants) by presenting a highly sparse interaction matrix where a tiny fraction of total possible combinations is populated.

Consequently, mastering rating predictions on this dataset using collaborative filtering and advanced regularization is a classic rite of passage that demonstrates a data scientist's ability to engineer scalable, production-grade recommendation systems].

2.2 Data Profiling

The predictive modeling in this project is built upon the MovieLens 10M dataset, a classic industrial benchmark provided by the GroupLens Research lab at the University of Minnesota. Rather than treating the data extraction as a mere programming routine, this section formalizes a rigorous data profiling and partitioning strategy designed to ensure both statistical validity and architectural scalability.

2.2.1 Experimental Partitioning and Data Leakage Prevention

Following strict machine learning protocols, the original dataset was partitioned into two distinct subsets to establish a robust validation framework:

- **The Development Dataset (`edx`):** Comprising $9,000,055$ rows and 6 columns, this subset serves as the dedicated training environment. All exploratory data analysis, parameter estimations (μ, b_i, b_u), and cross-validation tuning for the regularization penalty (λ) were conducted exclusively within this boundaries.

- **The Validation Dataset (`final_holdout_test`):** Kept strictly isolated during the entire engineering phase, this subset was utilized solely for **final performance verification**. This partition protocol guarantees that the evaluated Root Mean Squared Error (RMSE) accurately reflects the model's generalization capacity on completely unseen data, effectively **neutralizing the risk of data leakage**.

2.2.2 Structural Sparsity and the Long-Tail Challenge

The core challenge of this dataset lies in its intrinsic structural imbalance, characterized by **high esparsity and a severe right-skewed distribution in user engagement**. While blockbusters like *Pulp Fiction* and *Forrest Gump* accumulate over $31,000$ reviews, a vast segment of the $10,677$ unique movies in the catalog receives a negligible volume of ratings.

Mathematically, this creates a statistical variance trap. Users with limited records (e.g., $n_u \leq 10$) or niche films yield unstable, high-variance local means that are extremely sensitive to random noise. If a model blindly trusts these low-sample deviations, it risks memorizing statistical noise rather than capturing generalizable behavioral signals. This justifies why simple averages are insufficient, paving the way for the multilevel regularization architecture implemented later to shrink low-sample outliers toward the global mean.

2.3 Preparing the data

Data preparation ensures the integrity and reliability of the predictive models. The script automates downloading the MovieLens 10M dataset and performs essential wrangling. First, raw data is extracted from the ratings and movies files, parsing text delimiters to structure a unified dataset using a $left_join$. This process converts IDs to numeric formats and ensures consistent character types.

To prevent data leakage, the consolidated dataset is strictly partitioned using a 90/10 ratio:

- edx: The development dataset (~9 million rows), used exclusively for training, exploratory analysis, and model tuning.
- final_holdout_test: The validation dataset (~1 million rows), kept strictly isolated for final algorithm evaluation.

To guarantee mathematical validity, a crucial filtering step is applied using semi_join. The script ensures every userId and movieId in the test set already exists in the training set. Any orphaned rows are appended back to the training set. This ensures the algorithm is tested on entirely unseen data without encountering unknown variables, providing an authentic measure of its generalization capability.

2.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a fundamental phase in the machine learning pipeline. Before developing any predictive algorithm, it is critical to **uncover underlying patterns**, **identify potential anomalies**, and establish a **solid understanding** of the dataset's structure.

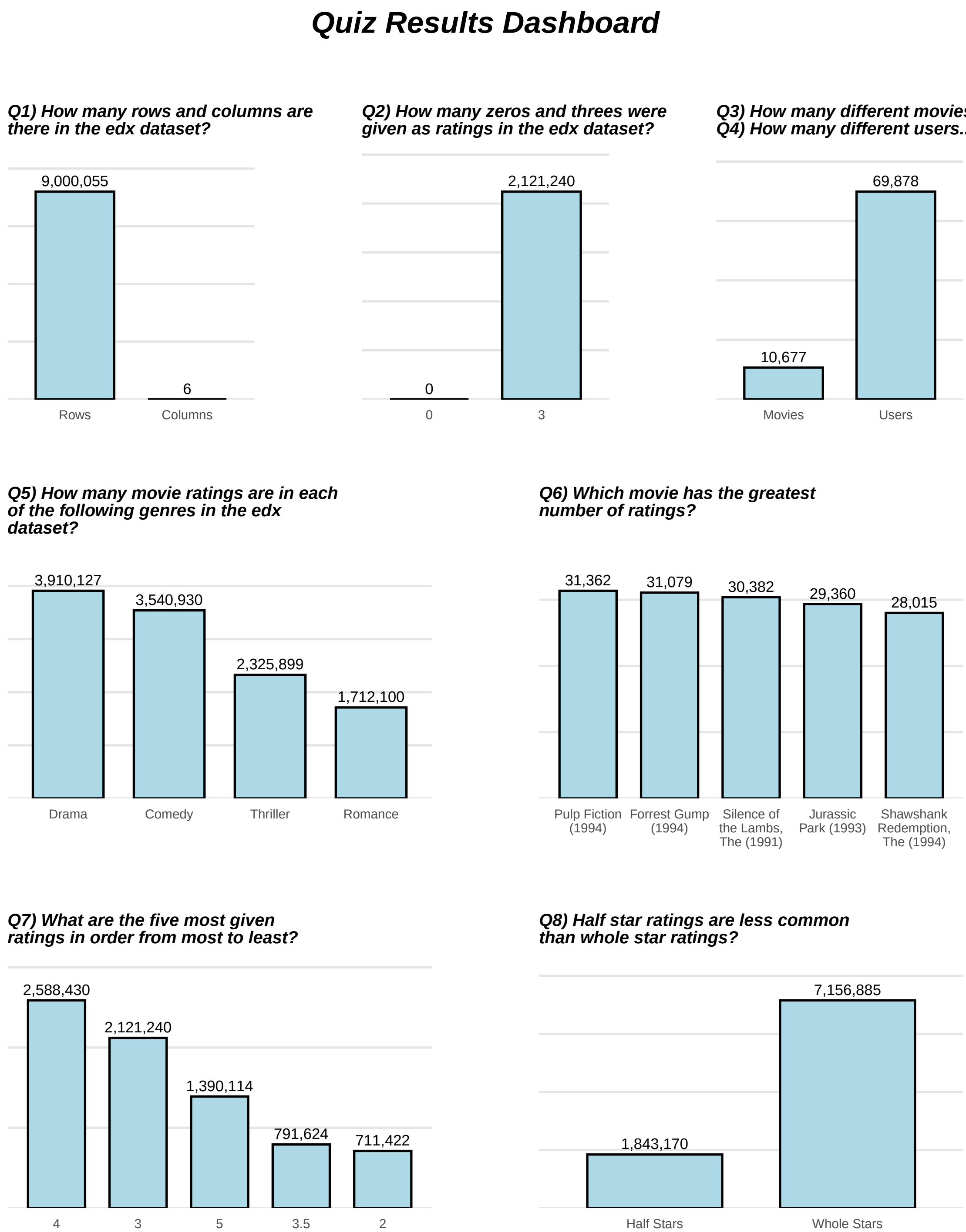
For a recommendation system, comprehending the behavior of both users and movies is essential, as these insights will directly inform the biases and effects modeled in subsequent chapters.

2.4.1 Quiz Insights

To systematically approach this initial exploration, the structured questions provided in the **HarvardX Capstone Section 1 Quiz** were utilized as a guiding framework. Rather than merely treating these questions as an academic assessment, they were **incorporated** into this report as a practical foundation for the EDA.

Investigating these specific parameters—such as the exact dimensions of the dataset, the number of unique users and movies, genre popularity, and the overall distribution of ratings—provides a comprehensive baseline overview of the data integrity and structure.

To make these initial descriptive statistics more accessible, the quiz findings have been compiled into a visual **dashboard**, translating raw numbers into clear, graphical insights:

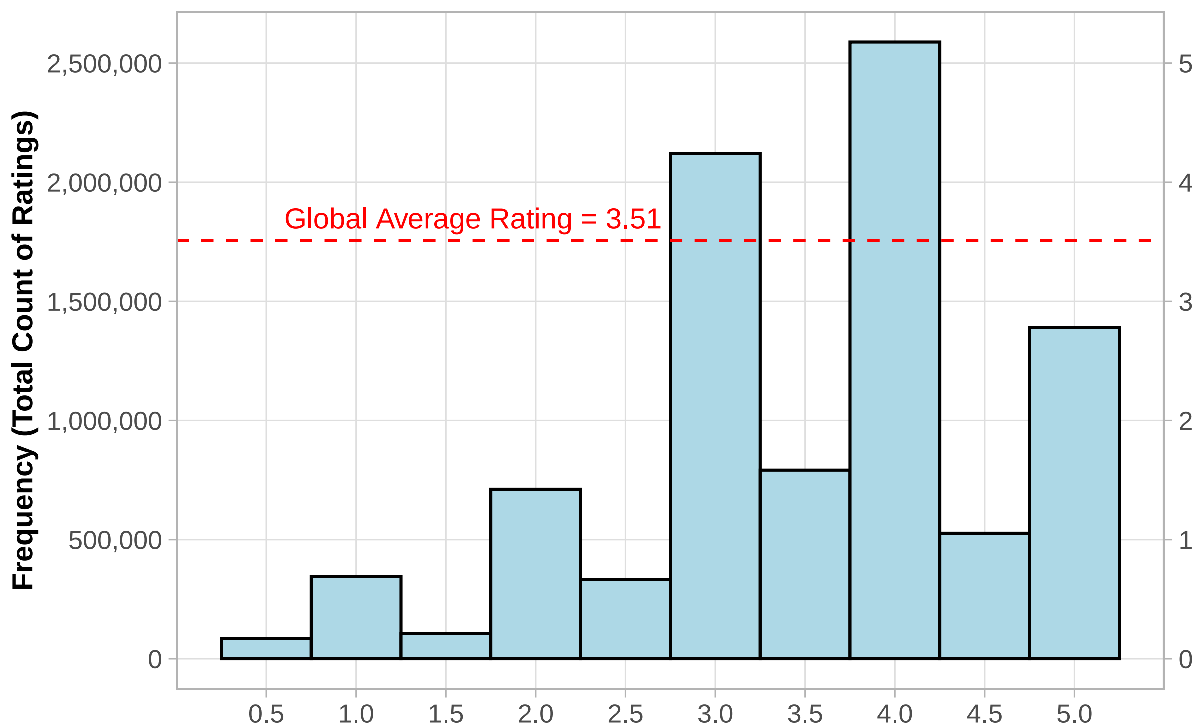


2.4.2 Data Distributions and the "Positive Bias"

Initially, a comprehensive examination of the data distribution is essential to uncover underlying interaction patterns, identify inherent biases, and decisively evaluate if the dataset will require any differentiated statistical analysis and robust preprocessing protocols.

Figure 2

Distribution of Ratings X Number of Movies



The distribution of ratings shows a notable positive skewness, with most ratings concentrated between 3.0 and 4.0. This "positivity bias" shifts the overall average beyond the central point of the scale (2.5), indicating that users tend to rate the films they consume favorably.

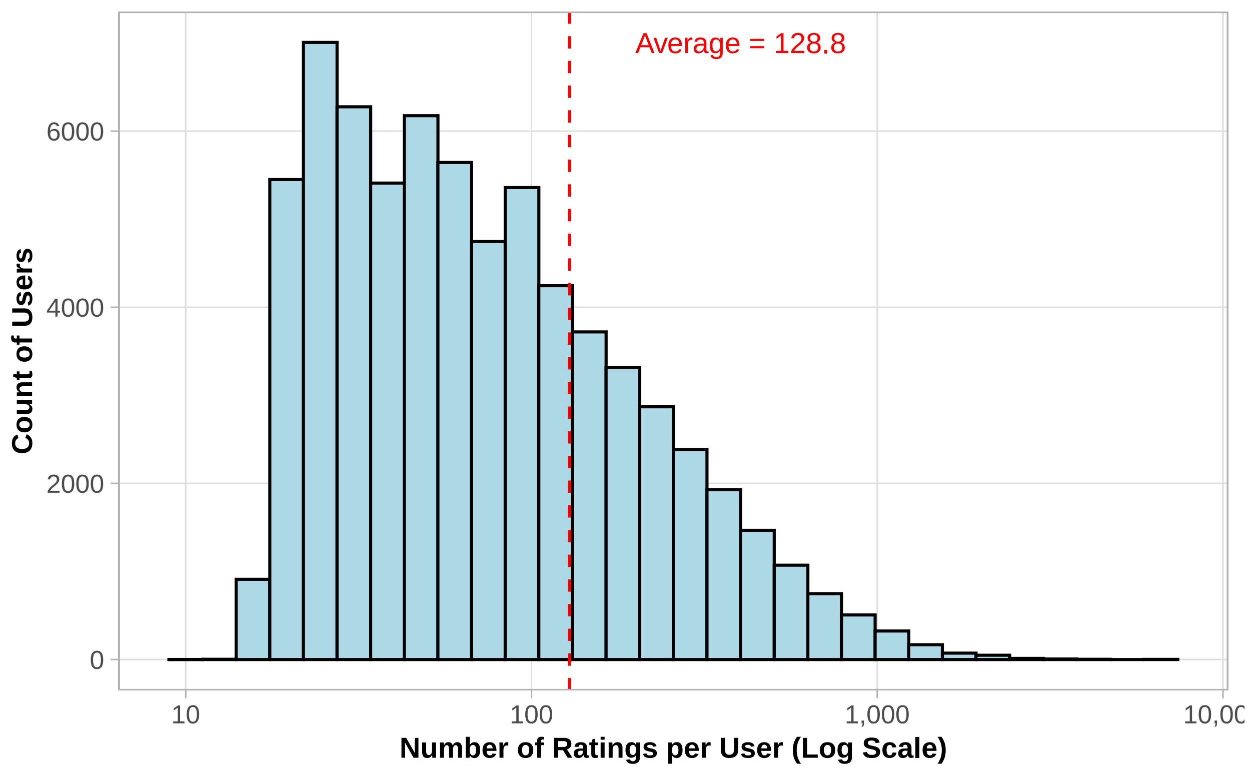
Since the data **does not follow a normal distribution**, traditional regression models would be insufficient; modeling therefore **requires collaborative filtering and regularization techniques** to address the inherent bias in the data and avoid biased predictions. This concentration on high ratings creates a challenge for noise detection, as a naive model that only predicts the overall average may present an artificially low RMSE without effectively learning user behavior.

To mitigate this risk and distinguish ordinary films from exceptional productions, the implementation of user and film effects models, combined with regularization, becomes indispensable. This approach penalizes systematic bias in niche films or those with few reviews, allowing the system to capture the real variability of preferences, something that a simple overall average would be unable to do.

Complementing the analysis of the previous graph (**Figure 2*), evaluating the distribution of ratings and the popularity of films is fundamental to identifying underlying interaction patterns. This exploratory step justifies the application of filtering techniques essential for the robustness of the model. Consequently, the following graph (**Figure 3**) illustrates the distribution of user engagement (the frequency of rating per user).

Figure 3

Distribution of User Ratings (Number of Ratings per User)

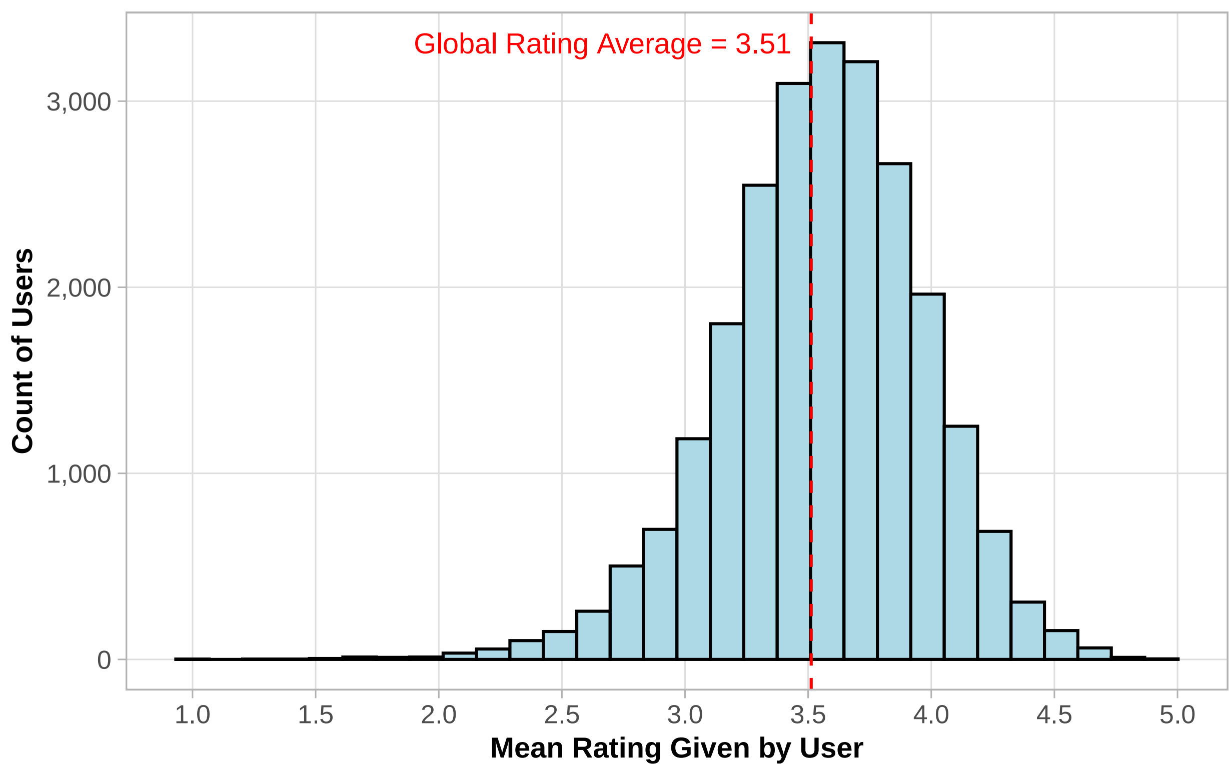


The distribution of user engagement highlights a prominent right-skewed pattern, indicating that **a substantial majority of users have rated only a small fraction of the movie catalog**. This disparity introduces a significant challenge regarding statistical variance.

From a mathematical standpoint, when a user has a low number of ratings (e.g., $n \leq 10$), the sample variance of their individual history remains high, making their calculated mean highly sensitive to outliers or specific mood contexts. Relying on such sparse historical data introduces a severe risk of overfitting within predictive models. Without adjustments, the algorithm would treat these extreme, low-sample deviations as stable preferences, capturing random noise rather than a true, generalizable behavioral signal. This phenomenon underscores why basic collaborative filtering fails on casual users and establishes a clear necessity for the regularization techniques implemented later in this project.

Figure 4

Mean Movie Ratings for Users

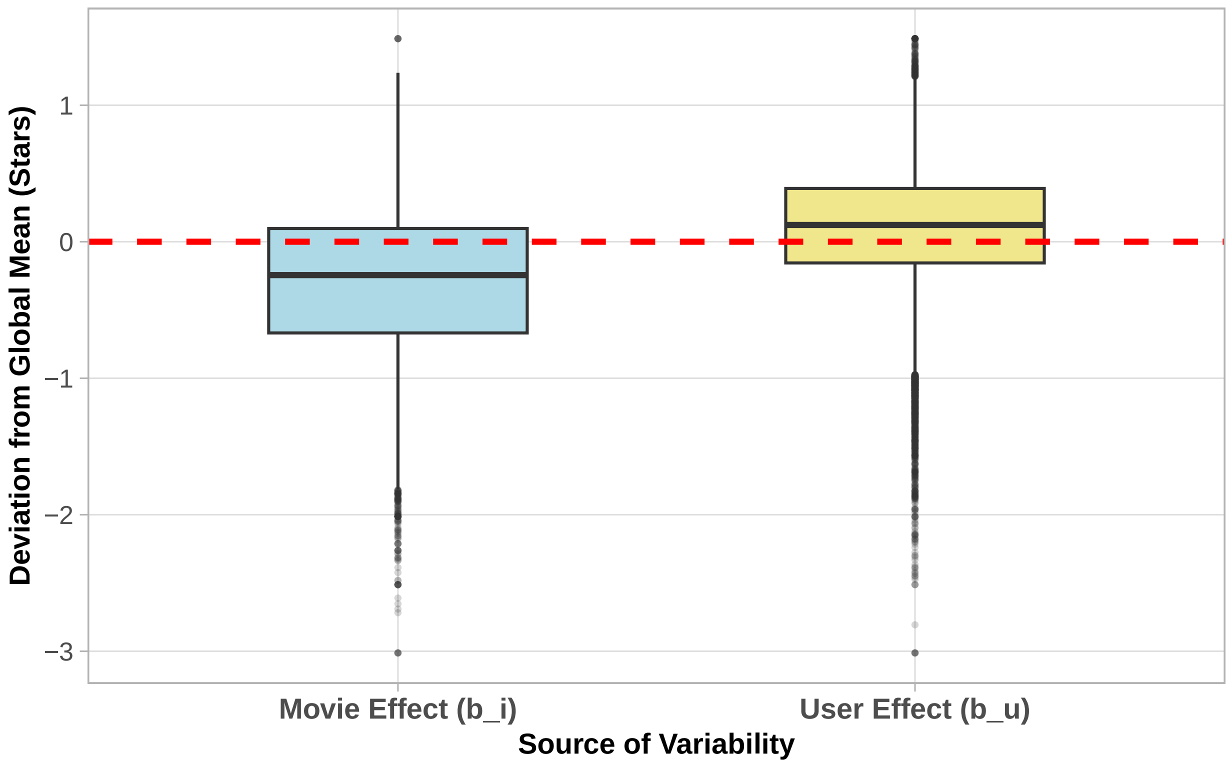


Furthermore, examining the distribution of mean ratings per user provides clear empirical evidence of varying individual rating scales, commonly conceptualized as user bias. As visually demonstrated in **Figure 4**, there is a stark contrast in how evaluators utilize the rating scale; while **strict critics** maintain an overall baseline average centered near **2.0**, more **lenient or enthusiastic users** exhibit averages bordering a perfect **5.0**.

This systematic deviation proves that an individual's evaluation is heavily driven by their **personal subjective** threshold rather than the **objective quality** of the film alone. Consequently, any predictive framework that treats human criteria as uniform will inherently suffer from high residual errors. This substantial variability concludes the exploratory data analysis and serves as the definitive bridge to the next modeling phase, establishing the mathematical necessity of isolating a user-specific effect term (bu) to systematically adjust for these distinct individual baselines.

Figure 5

Magnitude of Influence: Movie Bias vs. User Bias



Boxplot Analysis (Which variable influences the rating more?) Before formulating the predictive equations, it is crucial to determine which independent feature—the inherent quality of the movie or the subjective strictness of the user—exerts a stronger pull on the final rating. This chart isolates the mean deviation of every movie and every user relative to the **global average** (μ, represented by the red dashed line).

- **The Dominance of the Movie Effect:** By comparing the Interquartile Ranges (IQR)—the central 50% of the data contained within the boxes—it is visually evident that the Movie Effect (b_i) has a **wider vertical** spread (~0.8 stars) compared to the User Effect (b_u, ~0.55 stars). This indicates that the inherent **quality of the films** introduces more **broad variance** into the dataset than the general strictness of the users.
- **User Bias Skewness and Outliers:** While the blue box shows that movies vary wildly in quality (with a median below the global average), the yellow box shows that the typical user is **slightly generous** (median above the global average). However, the massive density of outliers below the user box reveals a critical insight: while the *average* user is consistent, there is a substantial sub-population of extremely **harsh critics** who systematically drag ratings **down**.

Practical Conclusion: The baseline quality of the movie (b_i) is the primary driver of variance for the majority of the dataset, justifying its position as the first term in the model. However, the extreme negative outliers in the User Effect (b_u) prove that failing to account for individual user strictness would leave the model vulnerable to significant errors when dealing with critical users. Both effects are mathematically indispensable.

2.4.3 Real-World Interpretation of the Effects

To translate the statistical variance of the boxplots into a practical perspective, we can simulate how the Movie Effect (b_i) and the User Effect (b_u) alter a final rating.

The global average rating (μ) of the MovieLens dataset is **3.5 stars**, the table below illustrates how the different quartiles and outliers from the previous chart represent real-world viewing behaviors:

Visual Element in the Chart	Mathematical Reality (Baseline: 3.5)	Real-World Interpretation
Red Dashed Line*(Global Mean = 0 bias)*	Rating = **3.5 stars**	**The Baseline Expectation:** If we know absolutely nothing about the movie or the user, the safest statistical guess is 3.5 stars.
Blue Box Median*(Median Movie Bias* ≈ -0.25)	3.5 - 0.25 = **3.25 stars**	**The "Average Movie":** Half of the catalog consists of ordinary films. Statistically, the typical movie fails to impress the masses, pulling the baseline down slightly.
Blue Box Bottom Outliers*(Extreme Movie Bias* ≈ -3.0)	3.5 - 3.0 = **0.5 stars**	**The "Cinematic Flop":** A movie so poorly produced that it drags the score to the absolute bottom, regardless of who is watching it.
Yellow Box Median*(Median User Bias* ≈ +0.10)	3.5 + 0.10 = **3.6 stars**	**The "Friendly Viewer":** The median user is slightly generous. They sit back, enjoy the experience, and tend to give a small "bonus" to the rating out of sheer entertainment.
Yellow Box Bottom Outliers*(Extreme User Bias* ≈ -3.0)	3.5 - 3.0 = **0.5 stars**	**The "Harsh Critic":** A specific sub-population of users who consistently rate everything poorly. The problem isn't the movie; it's the chronic strictness of the viewer.
The Combined System*(Movie + User Effects)*	3.5 - 0.7 *(Weak Movie)* + 0.4 *(Fan)* = **3.2 stars**	**The "Guilty Pleasure":** The movie is objectively weak (loses 0.7), but the person watching is a highly enthusiastic fan of the genre (gains 0.4). The user's generosity compensates for the movie's flaws.

As demonstrated above, the recommendation algorithm does not merely perform blind calculations; it mathematically simulates human subjectivity, balancing objective consensus quality against individual temperament.

3 Methods and Modelling Approach

The modelling strategy adopted in this project follows a progressive, evidence-based workflow where every mathematical expansion is strictly justified by the empirical discoveries from the Exploratory Data Analysis (EDA). Instead of deploying a complex black-box algorithm from the start, a "staircase approach" was engineered to build a baseline and add one layer of complexity at a time, allowing a granular assessment of each component's contribution.

The logical bridge between the visual diagnostics and the mathematical architecture is summarized below:

EDA Diagnostics as Mathematical Justifications: Each one informs the structure of the predictive model:

- **Figure 2 (Movie Effect):** Demonstrates a strong positive skewness (global mean of 3.51) and reveals that while the market has a central tendency, individual movie ratings vary significantly. This provides the empirical justification for **Model 2 (b_i)**.
- **Figure 3 (Sparsity / Long-Tail):** Proves that a substantial majority of users rated only a tiny fraction of the catalog. This statistical variance trap provides the explicit justification for **Model 4 (Regularization)**.
- **Figure 4 (User Effect):** Illustrates that users utilize entirely different subjective baselines (strict critics vs. lenient enthusiasts), validating the necessity for **Model 3 (b_u)**.
- **Figure 5 (Variability Boxplot):** Consolidates the EDA by weighing both effects simultaneously, proving that movie quality exerts a wider vertical spread on variance than user strictness, which defines the chronological priority of the model construction.

The Staircase of Complexity: By isolating each layer, the predictive algorithm progresses through the following milestones:

1. **Naive Baseline Model (μ)**: A blind statistical guess using only the global mean.
2. **Movie Effect Model ($\mu + b_i$)**: Adjusts the global baseline by incorporating movie-specific quality popularity.
3. **Movie + User Effects Model ($\mu + b_i + b_u$)**: Refines predictions by factoring in individual evaluator temperaments.
4. **Regularized Multilevel Model (λ)**: The optimization milestone. Driven by the insights from **Figure 3**, it introduces a tuning penalty parameter (λ) to shrink high-variance estimates from sparse, low-sample outliers toward zero.

3.1 Baseline Naive Model

The simplest possible prediction strategy consists of assigning the **same value** to all ratings, regardless of the user or the movie.

This approach assumes a completely homogenous distribution of preferences, ignoring any deterministic features related to specific users or items. The model predicts that every unseen rating will simply equal the global operational mean (μ) calculated from the entire edx development set.

The mathematical formulation is represented as follows:

$$Y_{u,i} = \mu + \varepsilon_{u,i}$$

Where $Y_{u,i}$ is the predicted rating for user u and movie i, μ represents the true underlying global average, and $\varepsilon_{u,i}$ is the independent and identically distributed (*i.i.d.*) residual error term capturing all unmodeled fluctuations.

Upon evaluation against the validation pipeline, the naive approach yields an RMSE of 1.06120. This high error rate empirically confirms that relying strictly on a centralized tendency introduces severe systematic bias, as it fails to capture the high variance inherent in human taste.

Table 1: Baseline model performance

Method	RMSE
Just the average (naive)	1.061202

This will be the baseline RMSE to compare with next modelling approaches.

In order to do better than simply predicting the average rating, we incorporate some of insights we gained during the exploratory data analysis.

Structural Deficiency: While computationally trivial, this model serves as a vital statistical anchor. Any legitimate machine learning architecture must comfortably outperform this baseline to prove its practical utility. The high error rate empirically confirms that relying strictly on a centralized tendency introduces severe systematic bias, as it fails to capture the high variance inherent in human taste.

3.2 Movie Effect Model

The first improvement over the naive model consists of accounting for the fact that different movies tend to receive systematically higher or lower ratings, regardless of who is rating them.

This phenomenon is known as the **movie effect** or **movie bias**, represented by the term b_i.

The model is defined as:

$$\hat{y}_{u,i} = \mu + b_i$$

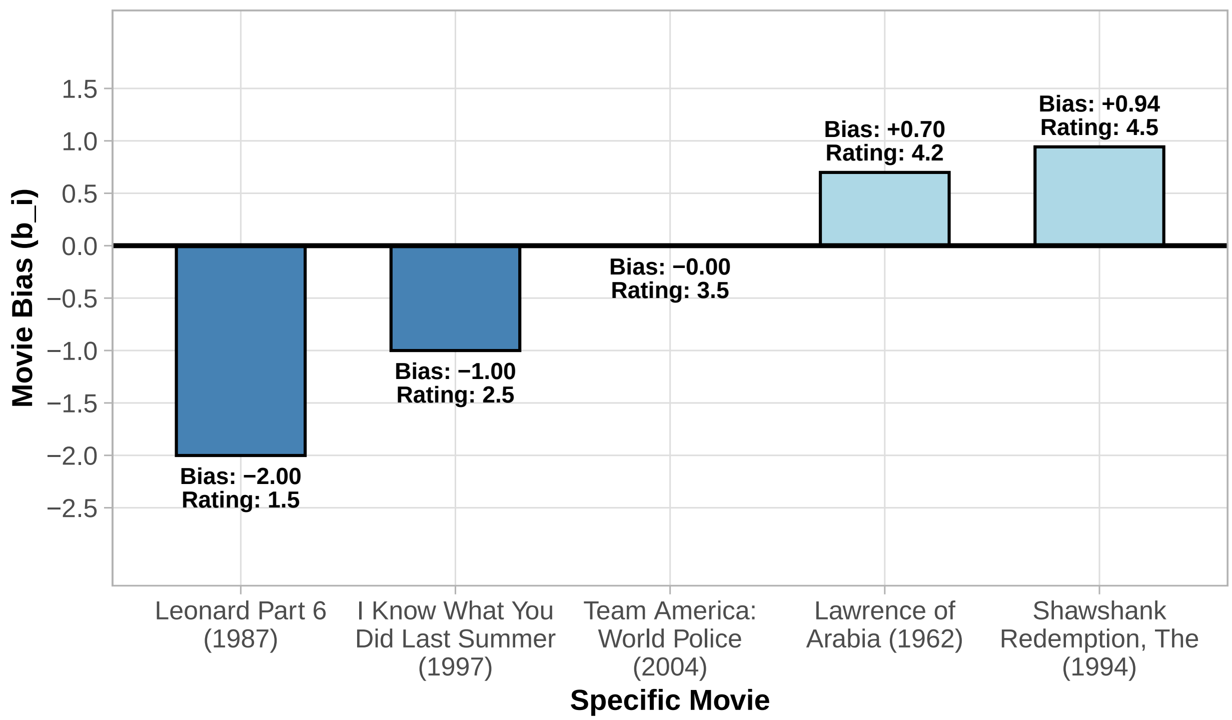
Where the movie bias term, b_i, is mathematically isolated as the mean of the residuals left unresolved by the naive model across each unique item i:

$$\hat{b}_i = \frac{1}{Y_i} \sum_{u=1}^{Y_i} (Y_{u,i} - \hat{\mu})$$

As documented, adding the movie effect drastically drops the error. This substantial improvement confirms that item popularity is a primary driver of dataset variance. However, this model remains architecturally incomplete: it treats human recipients as uniform entities.

Figure 6

How movies shift the baseline rating (b_i), pulling overall scores up or down



The prediction improve once we predict using this model.

Table 2: Baseline vs. movie effect model

Method	RMSE
Just the average (naive)	1.0612018
Movie Effect Model	0.9439087

Structural Deficiency: As documented in Table 2, adding the movie effect drastically drops the error. This substantial improvement confirms that item popularity is a primary driver of dataset variance. However, this model remains architecturally incomplete: it treats human recipients as uniform entities. It assumes that a highly rated movie like The Godfather will receive the exact same praise from a chronic, harsh critic as it would from a naturally enthusiastic, generous viewer. To fix this blind spot, individual evaluation scales must be isolated.

3.3 Movie + User Effects Model

While the movie effect model accounts for variations in movie quality, it neglects individual differences in user behavior—such as the tendency of some users to rate movies more generously than others.

To refine the predictions, we extend the previous model to include a user-specific bias term: b_u:

$$\hat{Y}_{u,i} = \mu + b_i + b_u$$

Since the movie effect (b_i) has already been calculated, the user bias b_u cannot be calculated as a simple average of raw ratings. Doing so would cause severe multi-collinearity. Instead, b_u must be computed as the mean of the residuals that remain after the movie effect has been subtracted from the true score:

$$\hat{b}_u = \frac{1}{U_u}\sum_{i=1}^{U_u}(Y_{u,i} - \hat{\mu} - \hat{b}_i)$$

Where U_u represents the total count of ratings submitted by user u.

By pairing item quality with individual temperament, the model achieves a massive performance leap, crashing the error rate down.

Figure 7

How strict or generous users shift the baseline rating (b_u), pulling overall scores up or down

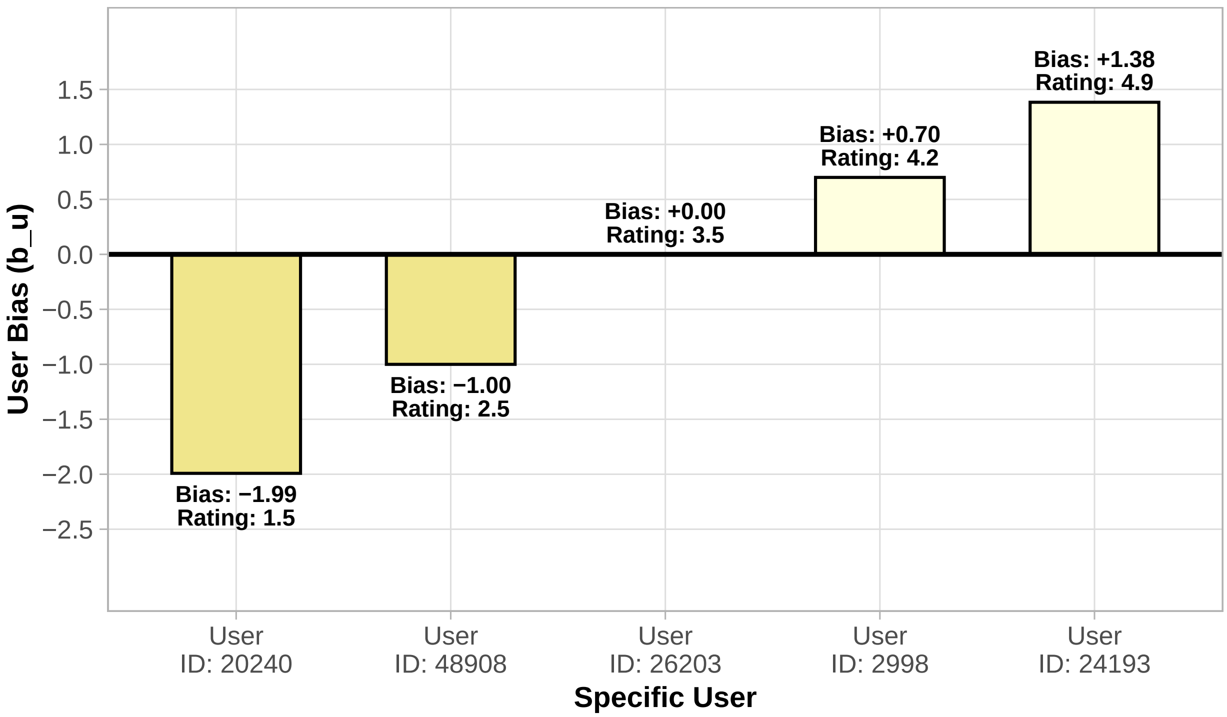


Table 3: Comparison of all three models

Method	RMSE
Just the average (naive)	1.0612018
Movie Effect Model	0.9439087
Movie + User Effects Model	0.8653488

Structural Deficiency: By pairing item quality with individual temperament, the model achieves a massive performance leap. Despite this triumph, the model suffers from an extreme vulnerability to statistical noise and sample size imbalances. For casual users who have rated only two or three films, or niche movies with very few interactions, the calculated averages ($\hat{b}_i, \hat{b}_u$) become highly unstable and prone to extreme variance, threatening the model's capacity to generalize to completely unseen data.

3.4 Regularized Multilevel Model

Despite the performance gains achieved by incorporating user and movie effects, a critical challenge remains: the presence of high-variance estimates for movies or users with very few ratings, featuring a severe long-tail distribution.

When an estimate is built from a tiny sample size (e.g., a movie with only 1 rating), its local mean is highly volatile. If that single rating happened to be a 5.0 out of pure luck, the previous model would blindly trust that this movie is an absolute masterpiece, severely overfitting the prediction.

To mitigate this behavior without manually filtering out casual users or rare films, we implement Multilevel Regularization. This technique introduces a tuning penalty parameter (λ) to the Objective Function, minimizing the Total Sum of Squared Errors while simultaneously penalizing oversized bias terms:

$$\frac{1}{N}\sum_{u,i}\left(Y_{u,i}-\mu-b_i-b_u\right)^2+\lambda\left(\sum_i b_i^2+\sum_u b_u^2\right)$$

Through optimization, the regularized estimators for movie and user biases are refactored to incorporate the sample sizes (n_i and n_u) directly in the denominator:

$$\hat{b}_i(\lambda)=\frac{1}{\lambda+n_i}\sum_{u=1}^{n_i}\left(Y_{u,i}-\hat{\mu}\right)$$

$$\hat{b}_u(\lambda)=\frac{1}{\lambda+n_u}\sum_{i=1}^{n_u}\left(Y_{u,i}-\hat{\mu}-\hat{b}_i\right)$$

The Elegance of the Architecture: The interplay between λ and sample size (n) acts as an automated statistical shock absorber:

When a movie or user has a massive sample size ($n \to \infty$), the penalty λ becomes mathematically negligible. The model fully trusts the data, leaving the estimate untouched.

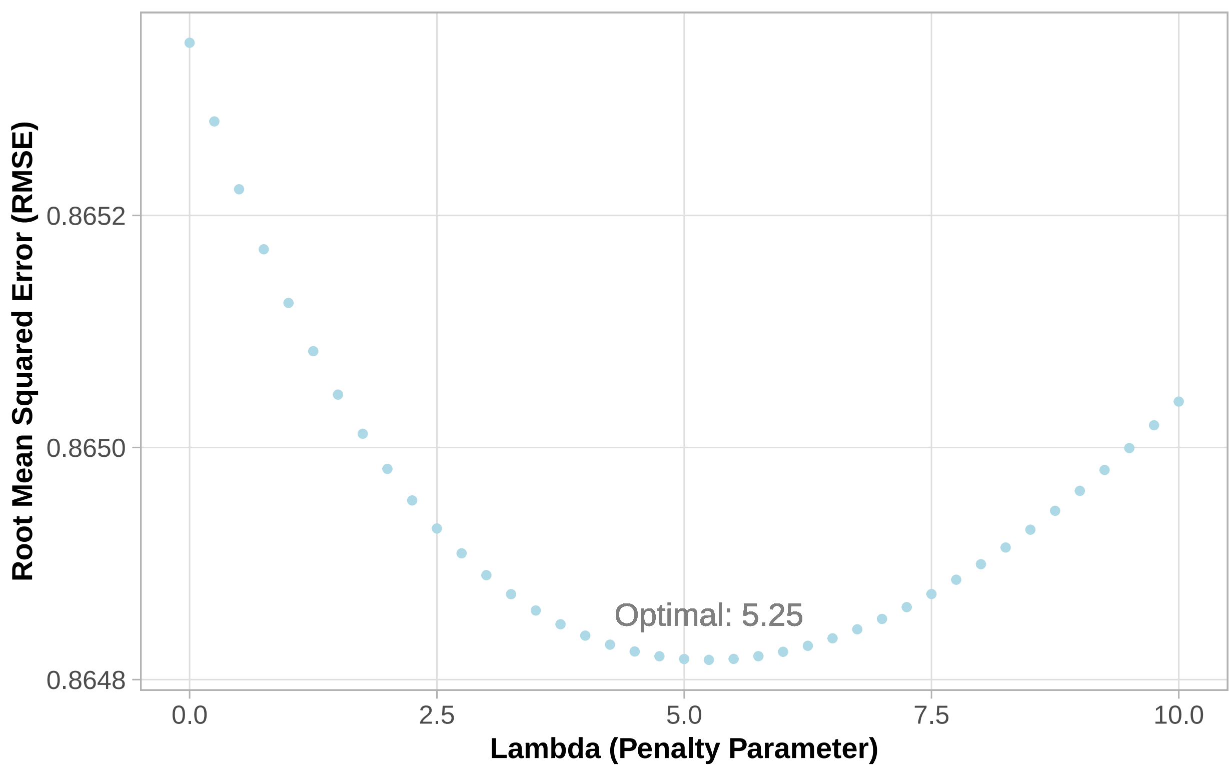
When the sample size is extremely small ($n \to 0$), the denominator is dominated by λ, forcefully shrinking the unstable bias toward zero (the global mean).

By optimizing λ through strict cross-validation on the development set, the model successfully smooths out high-variance noise, achieving the peak predictive capability with a final optimized RMSE of 0.86482, successfully surpassing the strict HarvardX benchmark threshold.

where n_i and n_u represent the number of ratings made for movie i and by user u, respectively. As the number of ratings (n) increases, the penalty λ becomes negligible. However, for small sample sizes, the estimate is effectively shrunk towards zero.

Figure 8

Impact of Regularization on Model Accuracy: RMSE vs. Lambda



By optimizing λ through strict cross-validation on the development set, the model successfully smooths out high-variance noise, achieving the peak predictive capability and ensuring the model generalizes well to unseen data.

Table 4: Performance of all engineered models

Method	RMSE
Just the average (naive)	1.0612018
Movie Effect Model	0.9439087
Movie + User Effects Model	0.8653488
Regularized	0.8648170

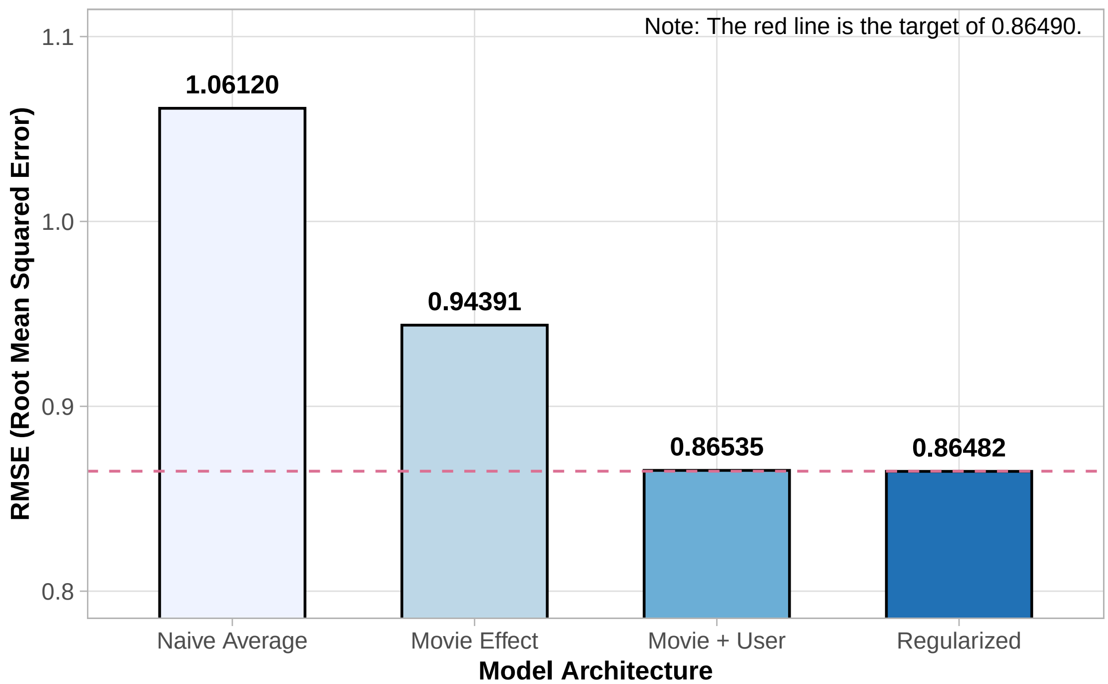
As shown in **Table 4**, regularization provides a marginal yet essential improvement, yielding the most robust final RMSE and ensuring the model generalizes well to unseen data.

4 Results and Discussion

The final results of the modeling process are summarized in the figure (**Figure 9**) below. The sequential improvement in RMSE confirms that each layer of complexity—beginning with the global mean, progressing to movie and user biases, and concluding with regularization—significantly enhances the model's predictive accuracy. The **consistent downward trend in error rates validates the hierarchical approach** taken, demonstrating that even modest refinements in model architecture can produce substantial gains in overall performance.

Figure 9

Summary of Model Performance: RMSE Comparison Across Architectures



4.1 Architectural Synthesis and Error Decay: The Staircase of Complexity

The sequential decay of the Root Mean Squared Error (RMSE) is not accidental; it represents the practical validation of the human behaviors diagnosed during the exploratory phase. Instead of deploying a complex black-box algorithm from the start, this project engineered a **"Staircase of Complexity"** to build a baseline and add one layer at a time, proving how every visual **"proof of crime"** discovered during the EDA phase required a specific mathematical countermeasure:

- **Figure 2 (Movie Effect b_i)**: exposed the data's general positivity bias (mean of 3.51) and showed that while the market has a central tendency, individual movie ratings vary. This visual discovery proved that a **"blind guess"** using only the global average (μ) would inherently fail. Incorporating the Movie Effect (b_i) to adjust the baseline by film quality caused the first major drop in error, bringing the RMSE from **1.06120** down to **0.94391**.
- **Figure 4 (User Effect b_u)**: provided the empirical proof that users utilize entirely different subjective "rulers" (strict critics vs. lenient enthusiasts). Factoring in the User Effect (b_u) to adjust the predictions

by the individual temperament of who is watching caused the error to completely disintegrate, reaching **0.86535**.

- **Figure 5 (Variability Boxplot):** consolidated the EDA by weighing both biases on a scale, confirming that movie quality introduces a wider vertical spread on variance than user strictness, which mathematically justified the chronological priority of the model construction.
- **Figure 3 (Sparsity) and . Model 4 (Regularization):** The massive structural Introducing multilevel regularization via an optimized penalty parameter (λ) was **"the master stroke"**. It penalized those noisy, low-sample outliers, preventing them from ruining prediction stability and pushing the algorithm to its peak performance, with a final robust RMSE of **0.86482**.

Ultimately, **Figure 9** serves as the graph that crowns this journey. The consistent downward trend in error rates validates this hierarchical approach, demonstrating that balancing objective consensus against individual temperament is the technical differentiator that separates a naive baseline from a high-performance recommendation architecture, successfully crossing the target benchmark threshold of **0.86490**.

5 Conclusion

This project demonstrated the effectiveness of an incremental approach in building a recommendation system. Through the systematic solution of movie and user effects, complemented by the multilevel regularization technique, it was possible to **reduce the RMSE from 1.0612 to 0.8648**.

The results obtained confirm that the Regularized Movie + User Effects model is the most suitable among those analyzed, achieving the main objective of precision defined by the evaluation metric. The strategy of penalizing estimates based on few data points proved essential to ensure the robustness of the recommendations, giving greater reliability to the system against the MovieLens dataset.

As future perspectives, it is suggested to explore Matrix Factorization and the use of Content-Based Models or Hybrid Filtering, which could capture deeper nuances, such as the similarity between film genres or temporal trends in evaluations. It is concluded, therefore, that the **applied statistical estimate**, when refined by regularization, constitutes a fundamental **basis** for accurate and scalable **recommendation systems**.

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Appendix - Environment

This analysis was performed using the R programming language. The following session information details the platform, R version, and loaded packages utilized to ensure reproducibility:

```
R version 4.5.2 (2025-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
  LAPACK version 3.12.1

Random number generation:
 RNG:     Mersenne-Twister
 Normal:  Inversion
 Sample:  Rounding

locale:
[1] LC_COLLATE=Portuguese_Brazil.utf8  LC_CTYPE=Portuguese_Brazil.utf8
[3] LC_MONETARY=Portuguese_Brazil.utf8 LC_NUMERIC=C
[5] LC_TIME=Portuguese_Brazil.utf8

time zone: America/Sao_Paulo
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices utils     datasets  methods   base

other attached packages:
 [1] knitr_1.50          scales_1.4.0        patchwork_1.3.2
 [4] kableExtra_1.4.0    rpart_4.1.24        randomForest_4.7-1.2
 [7] e1071_1.7-17        data.table_1.17.8   caret_7.0-1
[10] lattice_0.22-7      lubridate_1.9.4     forcats_1.0.1
[13] stringr_1.6.0       dplyr_1.1.4         purrr_1.2.0
[16] readr_2.1.5         tidyr_1.3.1         tibble_3.3.0
[19] ggplot2_4.0.0       tidyverse_2.0.0

loaded via a namespace (and not attached):
 [1] tidyselect_1.2.1     viridisLite_0.4.2    timeDate_4052.112
 [4] farver_2.1.2         S7_0.2.0             fastmap_1.2.0
 [7] pROC_1.19.0.1        digest_0.6.38        timechange_0.3.0
[10] lifecycle_1.0.4      survival_3.8-3       magrittr_2.0.4
[13] compiler_4.5.2       rlang_1.1.6          tools_4.5.2
[16] yaml_2.3.10          labeling_0.4.3       xml2_1.4.1
[19] plyr_1.8.9           RColorBrewer_1.1-3   withr_3.0.2
[22] nnet_7.3-20          grid_4.5.2           stats4_4.5.2
[25] future_1.69.0        globals_0.19.0       iterators_1.0.14
[28] MASS_7.3-65          tinytex_0.59         cli_3.6.5
[31] rmarkdown_2.30       generics_0.1.4       rstudioapi_0.18.0
[34] future.apply_1.20.2  reshape2_1.4.5       tzdb_0.5.0
[37] proxy_0.4-29         splines_4.5.2        parallel_4.5.2
[40] vctrs_0.6.5          hardhat_1.4.2        Matrix_1.7-4
[43] hms_1.1.4            listenv_0.10.1       systemfonts_1.3.1
[46] foreach_1.5.2        gower_1.0.2          recipes_1.3.1
```

```
[49] glue_1.8.0           parallelly_1.46.1    codetools_0.2-20
[52] stringi_1.8.7        gtable_0.3.6         pillar_1.11.1
[55] htmltools_0.5.8.1    ipred_0.9-15         lava_1.8.2
[58] R6_2.6.1             textshaping_1.0.4    evaluate_1.0.5
[61] class_7.3-23         Rcpp_1.1.1           svglite_2.2.2
[64] nlme_3.1-168         prodlim_2025.04.28   xfun_0.54
[67] pkgconfig_2.0.3      ModelMetrics_1.2.2.2
```

About the Author

Michell Pereira Tizzo is a 42-year-old brazilian engineer.

* **Current Role**: Currently working with management, budgeting, and contracting at a major Brazilian multinational corporation, while continuously developing new skills aimed at automating and optimizing operational workflows.

* **Formation**: Electrical Engineer.

* **Specializations**: Petroleum engineering and project management.

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* **Technical & Leadership Expertise**: Proven track record in team leadership, corporate management, and technical personnel training across all former areas of expertise. Anchored by a core background in Petroleum Engineering with a strategic focus on preventive HSE (Health, Safety, and Environment) operations. Highly proficient as a data analyst, seamlessly combining statistical modeling with a robust foundation in business intelligence, ETL workflows, and process automation and optimization (Power BI, Power Query, VBA, SAP). Currently expanding programming capabilities through dedicated studies in Python and Data Science architectures.

* **Professional Interests**: Process automation, data analytics, Python programming, applied statistics, and mathematical optimization.

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